

Study of carbohydrates, fats, oils and proteins in pure form and detection of their presence in the given food stuffs.

Aim: To detect the presence of carbohydrates, proteins or fats and oils in the given food stuff.

Theory: Food is a necessary material which must be supplied to the body for its normal and proper functioning. The essential constituents of food are:

1. Carbohydrates 2. Proteins 3. Lipids (fats and oils) 4. Minerals 5. Vitamins 6. Water etc.

Apparatus: Test tube, test tube stand, test tube holder, glass rod, water bath, etc.

Chemicals: Fehling's solution (A and B), Benedict's solution, Iodine solution (I_2), conc. HNO_3 , 10% NaOH, 1% $CuSO_4$ solution, chloroform, benzene, 1% Ninhydrin solution etc.

1. Carbohydrates:

Carbohydrates are polyhydroxy aldehydes or polyhydroxy ketones, their derivatives and the substances which yield them on hydrolysis. The carbohydrates which can reduce Tollen's reagent or Fehlings solution are called reducing sugars. For example glucose. Sucrose is a non reducing sugar. Some other examples of carbohydrates are starch, cellulose, maltose etc.

2. Proteins:

Proteins are complex, polymeric, nitrogenous organic compounds having high molecular mass, on hydrolysis give α -amino acids having peptide/amide linkage —CO—NH— in their molecules. **known as peptide linkage.** Proteins play vital role in living organism. For example heamoglobin present in blood, casein in milk, albumin in egg.

3. Fats and oils:

Lipids are defined as water insoluble organic biomolecules extracted from plants and animal cells by non-polar solvents. Chemically fats and oils are triesters of glycerol and higher fatty acids. At ordinary temperature oils are liquids while fats are solids. For example oil seeds, ground nuts, vanaspati ghee, butter, etc.

Foods Stuffs Required:

For Carbohydrates: Glucose powder, wheat flour, rice, biscuits, fruits, potato, starch etc.

For Proteins : Milk, egg albumin (white) etc.

For Oil/Fat : Ground nut powder, butter, linseed oil, cotton seed oil, vansapati ghee etc.

Procedure:

First grind the given food stuff in mortar with a pestle and make its paste. Alternatively boil food stuff with minimum quantity of water. Use this extract or solution directly for the tests.

Perform the tests given below for carbohydrates, proteins, fats and oil by taking a small amount of extract each time. Record the results in a tabular form.

(Note: Only one food stuff is given in practical examination.)

Aim: To detect the presence of carbohydrates or proteins or fats and oils in the given food stuff.
Food stuff for carbohydrates (Perform any two tests)

Sr. No.	Test	Observation	Inference
1	Solubility: Substance + Water		
2	Fehling Test: Little substance + 1 mL Fehling solution 'A' and 'B'. Heat the test tube on a water bath	Brown ppt	carbohydrate is present
3	Benedict's Test: Little substance + add 2 mL Benedict's reagent. Heat gently.	—	—

Result : The given sugar powder food stuff contains carbohydrates

Food stuff for protein (Perform any two tests)

Sr. No.	Test	Observation	Inference
1	Biuret Test: Substance + 10% NaOH + few drops of 1% CuSO ₄ solution, shake well	—	—
2	Xanthoprotic Test: Substance + few drops of conc. HNO ₃ . Shake a little and leave undisturbed for some time	—	—
3	Ninhydrin Test: Substance + NaOH solution + few drop of Ninhydrin reagent. Heat.	—	—

Result: The given.....food stuff contains protein.

Food stuff for fats and oils (Perform any two tests)

Sr. No.	Test	Observation	Inference
1	Solubility: a. Substance + Water	Immiscible in H ₂ O	oil is present
	b. Substance + chloroform	—	
2	Spot Test: Put a drop of substance on filter paper	Miscible	oil is present
3	Acrolein Test: Substance + few crystals of potassium bisulphate. Heat	—	

Result: The given oil food stuff contains fats or oils.

Remark and sign of teacher:

MCQ

Select the most appropriate answer from given alternatives of each sub question.

- Fats and oils are -----
 - triesters of glycerol and fatty acids
 - diesters of glycerol and fatty acids
 - glycerol and fatty acids
 - glycol and fatty acids
- are insoluble in water but soluble in acidic or alkaline solutions
 - fats
 - oils
 - proteins
 - carbohydrates
- Proteins contain ----- linkage.
 - polypeptide
 - polyester
 - polypropylene
 - polystyrene
- Glucose powder when heated with Fehling solution gives -----
 - blue ppt
 - red ppt
 - green ppt
 - black ppt

Short answer questions

1. What are Carbohydrates?

Ans. Carbohydrates are polyhydroxy aldehydes or polyhydroxy ketones, or compounds that yield them on hydrolysis.

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2. Write the name of chemical test, when protein in food stuff is treated with concentrated HNO_3 .

Ans. Xanthoproteic test

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3. Write structure of peptide linkage present in proteins.

Ans. $-\text{CO}-\text{NH}-$

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4. Mention Biuret test for proteins in short.

Ans. Substance containing protein is treated with 10% NaOH and a few drops of 1% CuSO_4 solution. On shaking, a violet (bluish-violet) colour is produced.

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5. Write an important difference between oils and fats.

Ans. Fats are solid at room temperature and are mainly triesters of saturated fatty acids.

Oils are liquid at room temperature and are mainly triesters of unsaturated fatty acids.

B. Identification of two anions (Acidic Radicals) from the given mixture of salts.

The following water soluble anions are included in the XII practical examination.

CO_3^{2-} (Carbonate), NO_3^- (Nitrate), NO_2^- (Nitrite), SO_4^{2-} (Sulphate), SO_3^{2-} (Sulphite), Cl^- (Chloride), PO_4^{3-} (Phosphate), Br^- (Bromide), I^- (Iodide), $\text{C}_2\text{O}_4^{2-}$ (Oxalate), CH_3COO^- (Acetate) and S^{2-} (Sulphide)

The analysis of Acidic radicals involve following sequence.

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|--|----------------------------------|
| A. Preliminary tests | B. Dry tests for acidic radicals |
| C. Preparation of the Original solution (O.S.) | D. Wet tests for acidic radicals |
| E. Confirmatory Tests (C.T.) | F. Result (with correct valency) |

A. Preliminary tests:

Test	Observation	Inference
Color	i. Blue/Bluish green or green	Chlorides, nitrates, sulphates, carbonates of Cu^{2+} , Cr^{2+} , Ni^{2+} may be present.
	ii. Yellow or reddish	Chlorides, nitrates, sulphates of Fe^{3+} may be present.
	iii. Pink or faint pink	Chloride, nitrates, sulphates of Co^{2+} may be present.
	iv. White/colourless	Chloride, nitrates, sulphates, carbonates of Ca^{2+} , Ba^{2+} , Sr^{2+} or Mg^{2+} may be present.
Nature	i. Crystalline (soluble in water)	Cl^- , Br^- , I^- , NO_2^- , SO_4^{2-} , CO_3^{2-} , CH_3COO^- or compounds of NH_4^+ or K^+ may be present.
	ii. Amorphous	Water insoluble salts containing CO_3^{2-} , S^{2-} may be present.
	iii. Hygroscopic/Deliquescence	Chlorides, nitrates of Cu^{2+} , Fe^{3+} , Cr^{3+} , Zn^{2+} , Cu^{2+} , Mg^{2+} may be present.
	(salt absorbs moisture)	

B. Dry Tests for Acidic radicals (Test tube must be dry for following tests)

Test	Observations	Inference
1. Heating in a dry test tube:- Take a small quantity of the mixture in a clean and dry test tube and heat it strongly in an oxidising flame (Blue flame) observe the changes taking place.	i. Colourless, odourless gas , which turns lime water milky.	CO_3^{2-} , $\text{C}_2\text{O}_4^{2-}$ may be present.
	ii. Coloured gases with pungent smell	
	a. Reddish brown gas which turns FeSO_4 solution black	NO_2^- , NO_3^- may be present.
	b. Greenish yellow gas which turns starch iodide paper blue	Cl^- may be present.
	c. Reddish brown gas which turns moist starch paper orange yellow	Br^- may be present.
	d. Dark violet fumes which turns starch paper blue	I^- may be present.

Select the most appropriate answer from given alternatives of each sub question.

- The anion does NOT gives positive test with dil. H_2SO_4 and conc. H_2SO_4 .
 a. Cl^- b. CO_3^{2-} c. CH_3COO^- d. SO_4^{2-} [Ans: d. SO_4^{2-}]
- When moisten oxalic acid is taken between the fingers, it smells like vinegar, indicates the presence of following anion
 a. sulphate b. nitrate c. acetate d. nitrite [Ans: c. acetate]
- Canary yellow ppt. of ammonium phosphomolybdate indicates the presence of radical
 a. Cl^- b. SO_4^{2-} c. PO_4^{3-} d. NO_3^- [Ans: c. PO_4^{3-}]
- Smell of rotten eggs of a substance with dil. H_2SO_4 indicates the presence of radical
 a. SO_3^{2-} b. SO_4^{2-} c. S^{2-} d. $\text{C}_2\text{O}_4^{2-}$ [Ans: c. S^{2-}]

Short answer questions

- Write the formula of the compound formed in brown ring test.

Ans. The compound formed in brown ring test is $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$.

- Write the name of the complex formed, when sodium nitroprusside is added to O.S. of mixture in the confirmatory test of sulphide ion.

Ans. Sodium pentacyanonitrosylferrate (II) sulphide, $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$.

- Write the names of anion detected with the help of dil. H_2SO_4 .

Ans. CO_3^{2-} (carbonate ion), NO_2^- (nitrite ion), S^{2-} (sulphide ion),
 SO_3^{2-} (sulphite ion).

- Write the names of anion detected with the help of conc. H_2SO_4 .

Ans. Cl^- (chloride), Br^- (bromide), I^- (iodide), NO_3^- (nitrate),
 CH_3COO^- (acetate ion).

- What is lime water and what happens on passing CO_2 gas through it?

Ans. Lime water is solution of calcium hydroxide in water which is alkaline and turns milky on passing CO_2 gas through it.

- Describe the layer test for iodide ions?

Ans. Take solution containing iodide ions and Cl_2 water and chloroform, shake well. Chloroform layer acquire violet colour.

- How do you test the presence of sulphide ion in dry test.

Ans. Mixture of salts is treated with dil. H_2SO_4 which evolves colourless gas with smell of rotten eggs (H_2S) which turns lead acetate paper black.

Remark and sign of teacher: